



**SIMATS**  
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## **CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM**

### **EXPERIMENT NO: 15**

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**Program:**

```
#include <stdio.h>
#include <math.h>
typedef struct
{
    int x, y;
} Point;
double distance(Point a, Point b)
{
    int dx = a.x - b.x;
    int dy = a.y - b.y;
    return sqrt(dx * dx + dy * dy);
}
void findClosestPair(Point p[], int n)
{
    double minDist = 999999;
    int i, j, p1 = 0, p2 = 0;
    for (i = 0; i < n - 1; i++)
    {
```

```

        for (j = i + 1; j < n; j++)
        {
            double d = distance(p[i], p[j]);
            if (d < minDist)
            {
                minDist = d;

                p1 = i;
                p2 = j;
            }
        }
    }

    printf("Closest Pair: (%d,%d)-(%d,%d)\n",
           p[p1].x, p[p1].y, p[p2].x, p[p2].y);
    printf("Distance = %.2f\n", minDist);
}

int main()
{
    Point points[] = {
        {1, 2},
        {4, 5},
        {7, 8},
        {3, 1}
    };

    int n = 4;

    printf("Name: Dharshana Sivakumar\n");

    printf("Closest Pair of Points\n\n");

    findClosestPair(points, n);

```

**return 0;**

**}**

**Output:**

| main.c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Run ▶ | Output                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------|
| <pre>1 // Point structure 2 3 int x, y; 4 } Point; 5 6 double distance(Point a, Point b) 7 { 8     int dx = a.x - b.x; 9     int dy = a.y - b.y; 10    return sqrt(dx * dx + dy * dy); 11 } 12 13 void findClosestPair(Point p[], int n) 14 { 15     double minDist = 999999; 16     int i, j, p1 = 0, p2 = 0; 17 18     for (i = 0; i &lt; n - 1; i++) 19     { 20         for (j = i + 1; j &lt; n; j++) 21         { 22             double d = distance(p[i], p[j]); 23 24             if (d &lt; minDist) 25             { 26                 minDist = d; 27                 p1 = i; 28                 p2 = j; 29             } 30         } 31     } 32 33     printf("Closest Pair: (%d,%d)-(%d,%d)\n", 34           p[p1].x, p[p1].y, p[p2].x, p[p2].y); 35 36     printf("Distance = %.2f\n", minDist); 37 } 38 39 int main() 40 { 41     Point points[] = { 42         {1, 2}, 43         {4, 5}, 44         {7, 8}, 45         {3, 1} 46     }; 47 48     int n = 4; 49 50     printf("Name: Dharshana Sivakumar\n"); 51     printf("Closest Pair of Points\n\n"); 52 53     findClosestPair(points, n); 54 55     return 0; 56 }</pre> |       | <p>Name: Dharshana Sivakumar<br/>Closest Pair of Points</p> <p>Closest Pair: (1,2)-(3,1)<br/>Distance = 2.24</p> |